

Causal Inference

Lecture 2

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based on Prof. Arvid Sjölander's slides

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- 1 Recap - Introduction to potential outcomes
- 2 Exchangeability
- 3 Randomized trials
- 4 Observational studies
- 5 Causal Model
- 6 Directed Acyclic Graphs (DAGs)
- 7 Motivating example
 - Structural equations and potential outcomes
- 8 Covariate selection in DAGs
- 9 Motivating example, revisited
- 10 Potential problems

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- Let Y_x be the outcome that we would observe, for a given subject, if the subject potentially received exposure level x
 - Y_0 is the outcome when not exposed
 - Y_1 is the outcome when exposed
- Y_0 and Y_1 are referred to as **potential outcomes**
- Ideally - **and very unrealistically** - we could observe both potential outcomes for any given subject

Subject	Y_0	Y_1
August	0	1
Selma	0	0
Fjodor	1	1

subject	Y_0	Y_1
August	0	1
Selma	0	0
Fjodor	1	1

- X has a causal effect on Y , for a given subject, if the potential outcomes Y_0 and Y_1 differ for this subject
 - for August, the exposure has an effect: $Y_0 \neq Y_1$
 - for Selma and Fjodor, the exposure has no effect; $Y_0 = Y_1$

- August is exposed ($X = 1$). Thus, for August
 - Y_1 is observed and equal to the factual outcome Y
 - Y_0 is unobserved, or **counterfactual**
- Selma and Fjodor are unexposed ($X = 0$). Thus, for Selma and Fjodor
 - Y_0 is observed and equal to the factual outcome Y
 - Y_1 is unobserved, or **counterfactual**

subject	X	Y	Y_0	Y_1
August	1	1	?	1
Selma	0	0	0	?
Fjodor	0	1	1	?

A fundamental problem of causation

- It is very difficult to say whether the exposure causes the outcome for a specific subject
 - because we cannot observe the same subject under two exposure levels simultaneously
- Fortunately, it is much easier to justify causal claims on population levels
 - e.g. 'if nobody would smoke, then the incidence of liver cancer would be 15% less then if everybody would smoke'

- $p(Y_x = 1)$ is the probability of the outcome, if **everybody** would receive exposure level x
- X has a population causal effect on Y if

$$p(Y_0 = 1) \neq p(Y_1 = 1)$$

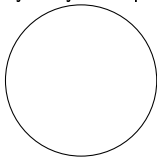
- X has no population causal effect on Y if

$$p(Y_0 = 1) = p(Y_1 = 1)$$

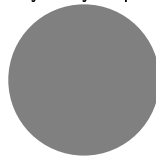
- Hereafter we refer to ‘population causal effects’ as ‘causal effects’

How to estimate population causal effects

Everybody unexposed



Everybody exposed



$$p(Y_0 = 1) \text{ vs } p(Y_1 = 1)$$

- From the definition we need to compare
 - the whole population being unexposed, with
 - the whole population being exposed
- But just like for any given subject, we cannot observe the whole population under two exposure levels simultaneously
- *How to solve this problem?*

A possible solution

Factually unexposed



Factually exposed



$$p(Y = 1|X = 0) \text{ vs } p(Y = 1|X = 1)$$

- We can use the associational risk ratio

$$\frac{p(Y = 1|X = 1)}{p(Y = 1|X = 0)}$$

as an estimate of the causal risk ratio

$$\frac{p(Y_1 = 1)}{p(Y_0 = 1)}$$

- When does this solution work?

- Intuitively, the associational risk ratio

$$\frac{p(Y = 1|X = 1)}{p(Y = 1|X = 0)}$$

is equal to the causal risk ratio

$$\frac{p(Y_1 = 1)}{p(Y_0 = 1)}$$

if there are no systematic differences between exposed and unexposed

- If so, we say that have **exchangeability** between exposed and unexposed
- The concept of exchangeability can be formalized with potential outcomes

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Example

ID	X	Y
1	0	0
2	0	0
3	0	1
4	0	1
5	0	0
6	1	0
7	1	1
8	1	1
9	1	1
10	1	0

- *Compute the associational risk ratio*

ID	X	Y
1	0	0
2	0	0
3	0	1
4	0	1
5	0	0
6	1	0
7	1	1
8	1	1
9	1	1
10	1	0

$$p(Y = 1|X = 0) = 2/5 = 0.4$$

$$p(Y = 1|X = 1) = 3/5 = 0.6$$

$$\frac{p(Y = 1|X = 1)}{p(Y = 1|X = 0)} = \frac{0.6}{0.4} = 1.5$$

- What we ultimately want is the causal risk ratio

$$\frac{p(Y_1 = 1)}{p(Y_0 = 1)},$$

not the associational risk ratio

$$\frac{p(Y = 1|X = 1)}{p(Y = 1|X = 0)}$$

- But perhaps the associational risk ratio is equal to the causal risk ratio?
- Let's see if we can re-write the associational risk ratio as the causal risk ratio
 - we will do this in two steps

ID	X	Y	Y_0	Y_1
1	0	0	0	?
2	0	0	0	?
3	0	1	1	?
4	0	1	1	?
5	0	0	0	?
6	1	0	?	0
7	1	1	?	1
8	1	1	?	1
9	1	1	?	1
10	1	0	?	0

- When $X = 0$ we have that $Y_0 = Y$, so that

$$p(Y = 1|X = 0) = p(Y_0 = 1|X = 0)$$

- When $X = 1$ we have that $Y_1 = Y$, so that

$$p(Y = 1|X = 1) = p(Y_1 = 1|X = 1)$$

- Putting together we have that

$$\frac{p(Y = 1|X = 1)}{p(Y = 1|X = 0)} = \frac{p(Y_1 = 1|X = 1)}{p(Y_0 = 1|X = 0)}$$

- Not home yet; the right-hand side is not a causal risk ratio. *Why?*

- The risk ratio

$$\frac{p(Y_1 = 1|X = 1)}{p(Y_0 = 1|X = 0)}$$

is not a causal risk ratio since it compares different groups

- exposed in the numerator, unexposed in the denominator
- Causal effects always compare potential outcomes for the same group (e.g. the whole population)

ID	X	Y	Y_0	Y_1
1	0	0	0	?
2	0	0	0	?
3	0	1	1	?
4	0	1	1	?
5	0	0	0	?
6	1	0	?	0
7	1	1	?	1
8	1	1	?	1
9	1	1	?	1
10	1	0	?	0

- Next, we want to have that

$$\frac{p(Y_1 = 1|X = 1)}{p(Y_0 = 1|X = 0)} = \frac{p(Y_1 = 1)}{p(Y_0 = 1)}$$

- This equality holds if

$$p(Y_0 = 1|X = 0) = p(Y_0 = 1)$$

and

$$p(Y_1 = 1|X = 1) = p(Y_1 = 1)$$

- Fill in values of Y_0 and Y_1 such that these equalities hold
- Verify that the associational risk ratio is equal to the causal risk ratio

- Several possible solutions, for instance:

ID	X	Y	Y_0	Y_1
1	0	0	0	0
2	0	0	0	1
3	0	1	1	1
4	0	1	1	1
5	0	0	0	0
6	1	0	0	0
7	1	1	0	1
8	1	1	1	1
9	1	1	1	1
10	1	0	0	0

$$p(Y_0 = 1|X = 0) = 2/5 = 0.4$$

$$p(Y_0 = 1) = 4/10 = 0.4$$

$$p(Y_1 = 1|X = 1) = 3/5 = 0.6$$

$$p(Y_1 = 1) = 6/10 = 0.6$$

$$\frac{p(Y_1 = 1)}{p(Y_0 = 1)} = \frac{0.6}{0.4} = 1.5$$

- Step 1:

1.

$$p(Y = 1|X = 0) = p(Y_0 = 1|X = 0)$$

$$p(Y = 1|X = 1) = p(Y_1 = 1|X = 1)$$

- This step follows from the definition of potential outcomes, and is always valid

- Step 2:

2.

$$p(Y_0 = 1|X = 0) = p(Y_0 = 1)$$

$$p(Y_1 = 1|X = 1) = p(Y_1 = 1)$$

- Is this step always valid?

Example

ID	X	Y	Y_0	Y_1
1	0	0	0	?
2	0	0	0	?
3	0	1	1	?
4	0	1	1	?
5	0	0	0	?
6	1	0	?	0
7	1	1	?	1
8	1	1	?	1
9	1	1	?	1
10	1	0	?	0

$$p(Y_0 = 1|X = 0) = p(Y_0 = 1)$$

$$p(Y_1 = 1|X = 1) = p(Y_1 = 1)$$

- Can you fill in values of Y_0 and Y_1 such that these equalities **do not** hold?

- Several possible solutions, for instance:

ID	X	Y	Y_0	Y_1
1	0	0	0	0
2	0	0	0	0
3	0	1	1	0
4	0	1	1	0
5	0	0	0	0
6	1	0	1	0
7	1	1	1	1
8	1	1	1	1
9	1	1	1	1
10	1	0	1	0

$$p(Y_0 = 1|X = 0) = 2/5 = 0.4$$

$$p(Y_0 = 1) = 7/10 = 0.7$$

$$p(Y_1 = 1|X = 1) = 3/5 = 0.6$$

$$p(Y_1 = 1) = 3/10 = 0.3$$

$$\frac{p(Y_1 = 1)}{p(Y_0 = 1)} = 0.43$$

$$\frac{p(Y = 1|X = 1)}{p(Y = 1|X = 0)} = 1.5$$

- Severe bias!

2.

$$p(Y_0 = 1|X = 0) = p(Y_0 = 1)$$

$$p(Y_1 = 1|X = 1) = p(Y_1 = 1)$$

- Clearly, it is possible that these equalities do not hold, in which case the second step is not valid
- As a consequence, association $\neq$ causation
- We will discuss when and why these equalities may hold, and when they may be violated
- Before doing so we will put a name on these equalities

$$\begin{aligned}p(Y_0 = 1|X = 0) &= p(Y_0 = 1) \\p(Y_1 = 1|X = 1) &= p(Y_1 = 1)\end{aligned}$$

- If these equalities hold, then we say that we have **exchangeability**
- If we have exchangeability, then association = causation

Alternative formulation of exchangeability

ID	X	Y	Y_0	Y_1
1	0	0	0	0
2	0	0	0	1
3	0	1	1	1
4	0	1	1	1
5	0	0	0	0
6	1	0	0	0
7	1	1	0	1
8	1	1	1	1
9	1	1	1	1
10	1	0	0	0

- The probability of the potential outcome Y_0 is the same among the unexposed as among everybody:

$$p(Y_0 = 1|X = 0) = p(Y_0 = 1)$$

- The probability of the potential outcome Y_1 is the same among the exposed as among everybody:

$$p(Y_1 = 1|X = 1) = p(Y_1 = 1)$$

- Exchangeability means that the potential outcomes (Y_0, Y_1) are independent of the exposure

$$(Y_0, Y_1) \perp\!\!\!\perp X$$

ID	X	Y	Y_0	Y_1
1	0	0	0	?
2	0	0	0	?
3	0	1	1	?
4	0	1	1	?
5	0	0	0	?
6	1	0	?	0
7	1	1	?	1
8	1	1	?	1
9	1	1	?	1
10	1	0	?	0

$$p(Y_0 = 1|X = 0) = p(Y_0 = 1)$$

$$p(Y_1 = 1|X = 1) = p(Y_1 = 1)$$

$$(Y_0, Y_1) \perp\!\!\!\perp X$$

- *Can we use the observed data to test if we have exchangeability or not?*

ID	X	Y	Y_0	Y_1
1	0	0	0	?
2	0	0	0	?
3	0	1	1	?
4	0	1	1	?
5	0	0	0	?
6	1	0	?	0
7	1	1	?	1
8	1	1	?	1
9	1	1	?	1
10	1	0	?	0

$$p(Y_0 = 1|X = 0) = p(Y_0 = 1)$$

$$p(Y_1 = 1|X = 1) = p(Y_1 = 1)$$

$$(Y_0, Y_1) \perp\!\!\!\perp X$$

- To test for exchangeability we need to have both potential outcomes (Y_0, Y_1) for all subjects
- In practice we only have one, so **exchangeability is untestable**

- We could only test for exchangeability if we would have both potential outcomes for all subjects
- But if we (very hypothetically) would have both potential outcomes, then we could directly compute the causal effect, even if we don't have exchangeability
- Thus, the only scenario where we could test for exchangeability is precisely the scenario where we don't need it

- In a randomized trial we assign exposure levels randomly to each study participant
 - e.g. by the flip of a coin
- **This study design guarantees that we have exchangeability**
- ... so that association = causation

- Under randomization, all pre-exposure variables are independent of the exposure
 - e.g. same distribution of age, sex and genetics in exposed and unexposed
- **The potential outcomes (Y_0, Y_1) are pre-exposure variables**
 - they describe how the subject will react to $X = 0$ and $X = 1$
 - this depends on factors that are determined before the factual exposure level is received. e.g. age, sex and genetics
- Thus, under randomization (Y_0, Y_1) are independent of X

$$(Y_0, Y_1) \perp\!\!\!\perp X$$

- This is amazing! Why then not always randomize?

- Research question: does obesity (X) increase the risk of cardiovascular disease (Y)?
- To answer this question with a randomized trial we must:
 - get funding and ethical approval
 - collect study participants
 - randomize to different levels of BMI
 - follow for, say, 5 years
 - compare the incidence of cardiovascular disease across BMI levels
- *Is this feasible?*

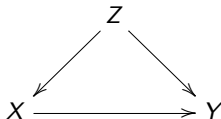
- We know today that obesity causes many different health problems
- Thus, we would most likely not get ethical approval for a study where the participants are randomized to obesity

- To make sure that the participants reach their assigned BMI levels, we must be able to control BMI by intervention
- In principle, we can think about possible interventions such as exercise, diet etc
- In practice, there is no guarantee that these interventions would work well (e.g. give the desired BMI)

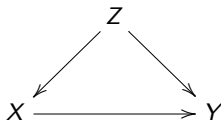
- Some participants may not comply with the study protocol
- The study is no longer an 'ideal' randomized trial - exchangeability is not guaranteed
- Traditional analyses:
 - Intention To Treat (ITT)
 - As Treated (AT)
- Both these analyses are likely to be biased
 - alternative 'instrumental variables methods' exist (later)

- In principle, randomized trials are ideal from a causal inference perspective
- However, randomization is often not possible for ethical and practical reasons
- Furthermore, real randomized trials often suffer from non-compliance, and exchangeability is not guaranteed
- We need observational studies!
 - in fact, most scientific knowledge comes from observations, e.g. evolution theory, smoking causes lung cancer etc
- ... and we thus also need methods for causal inference from observational studies

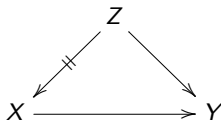
- In observational studies, the exposure is not controlled by design
- **The study design does not guarantee that we have exchangeability**



- Suppose that there is a covariate, Z , which affects both X and Y
 - e.g. $Z = \text{age}$; old people have higher BMI (X) than young people, and are at higher risk for cardiovascular disease (Y)
- If so, then X and Y will be associated, even in the complete absence of a causal effect
 - e.g. those with high BMI are older than those with low BMI, and for that reason at higher risk of cardiovascular disease
- Thus, association $\neq$ causation, i.e. we don't have exchangeability



- When X and Y have common causes, we say that there is **confounding**
- The consequence of confounding is that we don't have exchangeability
- The common causes are called **confounders**
 - we will modify this definition of 'confounder' later



- When X is randomized the influence of all confounders is broken by design
- Thus, there is no confounding in an ideal randomized trial

How can we estimate causal effects when there is confounding?

- We usually try to control/adjust for potential confounders in the statistical analysis
 - stratification
 - matching
 - standardization
 - propensity scores
 - regression modeling
 - inverse probability weighting
 - etc

- All methods to control for covariates Z aim to achieve conditional exchangeability, given Z :

$$(Y_0, Y_1) \perp\!\!\!\perp X \mid Z$$

- we say that Z is sufficient for confounding control
- Under conditional exchangeability, given Z , controlling for Z gives causal effects
- Conditional exchangeability is untestable
- Exchangeability can be achieved by controlling for covariates, but can also be 'destroyed'
 - more later

Example

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

- Assume conditional exchangeability, given Z , and compute the conditional causal risk ratio, given $Z = 0$ and $Z = 1$
- Where in the computation do you use the assumption of conditional exchangeability?

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

Conditional causal risk ratio, given $Z = 0$:

$$\begin{aligned}
 & \frac{p(Y_1 = 1|Z = 0)}{p(Y_0 = 1|Z = 0)} \stackrel{(Y_0, Y_1) \perp\!\!\!\perp X|Z}{=} \frac{p(Y_1 = 1|X = 1, Z = 0)}{p(Y_0 = 1|X = 0, Z = 0)} \\
 &= \frac{p(Y = 1|X = 1, Z = 0)}{p(Y = 1|X = 0, Z = 0)} \\
 &= \frac{20/200}{30/300} = 1
 \end{aligned}$$

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

Conditional causal risk ratio, given $Z = 1$:

$$\begin{aligned}
 & \frac{p(Y_1 = 1|Z = 1)}{p(Y_0 = 1|Z = 1)} \stackrel{(Y_0, Y_1) \perp\!\!\!\perp X|Z}{=} \frac{p(Y_1 = 1|X = 1, Z = 1)}{p(Y_0 = 1|X = 0, Z = 1)} \\
 &= \frac{p(Y = 1|X = 1, Z = 1)}{p(Y = 1|X = 0, Z = 1)} \\
 &= \frac{240/300}{80/200} = 2
 \end{aligned}$$

- Given $Z = 0$, the conditional causal risk ratio is equal to 1 (no effect)
- Given $Z = 1$, the conditional causal risk ratio is equal to 2
- Effect modification by (interaction with) Z

- Exchangeability is defined as

$$(Y_0, Y_1) \perp\!\!\!\perp X$$

- Under exchangeability, the crude (without controlling for covariates) association is equal to a causal effect
- In ideal randomized trials, exchangeability follows by design
- In observational studies, we typically don't have exchangeability because of confounding
- Conditional exchangeability, given Z , is defined as

$$(Y_0, Y_1) \perp\!\!\!\perp X \mid Z$$

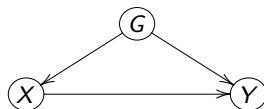
- Under conditional exchangeability, given Z , controlling for Z gives a causal effect
 - we say that Z is sufficient for confounding control

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What is a causal model?

- A causal model involves counterfactuals variables (and statements).
- A causal model can be mapped (represented) with a causal graph (DAG).
- A causal model allows us to answer questions about the effect of interventions.

DAG: The effect of the Treatment X on the outcome Y is confounded by the gender G .

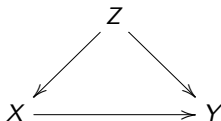


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- In ideal randomized trials we have exchangeability:

$$(Y_0, Y_1) \perp\!\!\!\perp X$$

- As a consequence, association = causation



- In observational studies we usually don't have exchangeability, because of confounding
- We may achieve conditional exchangeability by controlling for an appropriate set of covariates:

$$(Y_0, Y_1) \perp\!\!\!\perp X \mid Z$$

- Selecting an appropriate set of covariates to control for is a non-trivial task
- Exchangeability can be achieved by controlling for covariates, but can also be 'destroyed'

- Does smoking during pregnancy (exposure) cause malformation (outcome) in the child?
- We collect data on both exposure and outcome for a large number of pregnancies
- We measure five additional covariates:
 - the mother's age
 - the mother's socioeconomic status
 - the mother's diet during pregnancy
 - information on whether there is a family history of birth defects
 - information on whether the child was liveborn or stillborn

- We observe a negative association between smoking and malformation; risk ratio = 0.8
- However, we suspect that there is confounding of the exposure and outcome
 - then we don't have exchangeability, and the observed (associational) risk ratio is not equal to the causal risk ratio
- To eliminate, or at least reduce, confounding bias we want to control for measured covariates

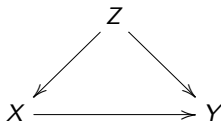
- One strategy would be to control for all measured covariates
- This strategy may not be optimal, because
 - **some covariates may not be confounders, and may increase bias (non-exchangeability) if controlled for**
 - more covariates requires a bigger model, with a higher potential for bias due to model misspecification
 - some covariates may be prone to measurement errors, and may therefore lead to bias
 - some covariates may reduce statistical power/efficiency when controlled for
- Therefore, it is often desirable to control for a subset of covariates

- Stepwise regression: select covariates that are strongly associated with the outcome
- 'Change-in-estimate': select covariates that substantially change the estimate of interest

- Require that data is already collected, and cannot not be used at the design stage
- May select non-confounders, which may increase bias (non-exchangeability) if controlled for

- Directed Acyclic Graphs (DAGs) can be used for knowledge-driven covariates selection
- A DAG is a graphical representation of underlying causal structures
- DAGs for covariate selection:
 - encode our *a priori* causal knowledge/beliefs into a DAG
 - apply simple graphical rules to determine what covariates to control for. The set of these rules is also called directional separation (d-separation) rules.
 - d-separation is a graphical criterion used in DAGs to determine conditional independence between two variables.
 - d-separation translates causal assumptions into testable conditional independence relationships.

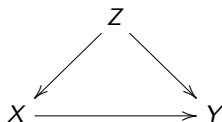
A simple DAG



- Each arrow represents a causal effect
- The graph is
 - Directed, since each connection between two variables consists of an arrow
 - Acyclic, since the graph contains no directed cycles
- Formal connection to potential outcomes/counterfactuals through non-parametric structural equations

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Structural Equations in NPSEMs:



- NPSEMs describe how variables are related via functions and unobserved noise.

- Example:

$$Z = f_1(\epsilon_Z), \quad X = f_2(Z, \epsilon_X), \quad Y = f_3(X, Z, \epsilon_Y)$$

- ϵ_X , ϵ_Z , and ϵ_Y are noise terms accounting for unobserved factors.

Definition:

Potential outcomes describe what the value of an outcome variable Y would be under different hypothetical interventions on a treatment variable X . For example:

$$Y_x = \text{the value of } Y \text{ if } X \text{ were set to } x.$$

Connection to Structural Equations:

- The potential outcome Y_x can be derived from the structural equations by intervening to set $X = x$ and removing all other dependencies of X .
- This is known as the do-operation in Pearl's framework.

Formally:

$$Y_x = f_3(x, Z, \epsilon_Y)$$

Here, X is replaced with x , and the value of Y_x is computed using the structural equations.

For the following NPSEM:

$$Z = \epsilon_Z, \quad X = f_1(Z, \epsilon_X), \quad Y = f_2(X, Z, \epsilon_Y)$$

To compute the potential outcome Y_x (when X is set to x):

- Replace X in the equation for Y with x .
- Compute Y as:

$$Y_x = f_2(x, Z, \epsilon_Y)$$

where Z and ϵ_Y remain unchanged.

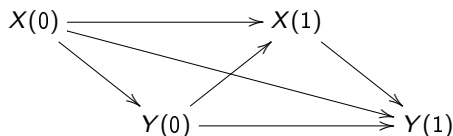
- The structural equation $X = f_1(Z, \epsilon_X)$ for X becomes irrelevant because we no longer allow X to vary based on its usual causes.
- This shows how structural equations directly define potential outcomes by encoding the causal system.

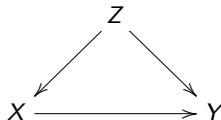
Key Points:

- NPSEMs explicitly describe causal mechanisms using structural equations.
- Directed Acyclic Graphs (DAGs) visually represent causal relationships.
- Potential outcomes are derived from structural equations via interventions.
- Counterfactuals provide insight into causal effects by comparing outcomes under different interventions.

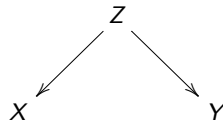
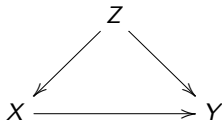
A note on acyclicity

- We impose acyclicity since a variable can't cause itself
 - e.g. my BMI today has no effect on my BMI today
- Observed variables are often snapshots of time varying processes
 - e.g. my BMI today certainly affects my BMI tomorrow
- Time varying processes can be depicted on a DAG by having one measure of each variable per time unit

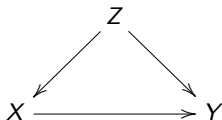




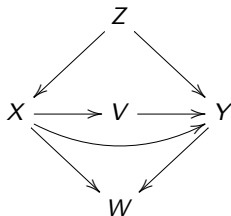
- Assumptions are encoded by the direction of arrows
 - the arrow from X to Y means that X may affect Y , but not the other way around



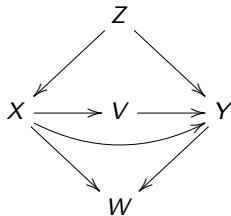
- Assumptions are encoded by the absence of arrows
 - the presence of an arrow from X to Y means that X may or may not affect Y
 - the absence of an arrow from X to Y means that X does not affect Y



- Assumptions are encoded by the absence of common causes
 - the presence of Z means that X and Y may or may not have common causes
 - the absence of Z means that X and Y do not have any common causes

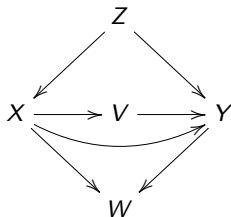


- A path is a route between two variables, not necessarily following the direction of arrows
- Which are the paths between X and Y ?

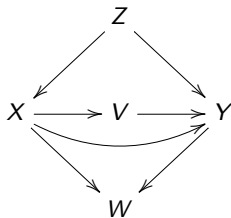


• Four paths between X and Y :

- $X \rightarrow Y$
- $X \rightarrow V \rightarrow Y$
- $X \leftarrow Z \rightarrow Y$
- $X \rightarrow W \leftarrow Y$

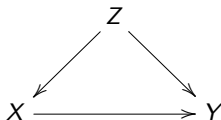


- A causal path is a route between two variables, **following the direction of arrows**
 - the causal paths from X to Y mediate the causal effect of X on Y , the non-causal paths do not
- *Which are the causal paths between X and Y ?*



- Two causal paths from X to Y :

- $X \rightarrow Y$
- $X \rightarrow V \rightarrow Y$



- Paths (causal and non-causal) are either open or blocked, depending on what variables we have controlled for
- Two nodes X and Y are **d-separated** (conditionally independent) given a set of nodes S if every path between X and Y is **blocked** by S .
- This is determined by two rules

- A path is blocked if somewhere along the path there is a variable Z that sits in a 'chain'

$$X \longrightarrow Z \longrightarrow Y$$

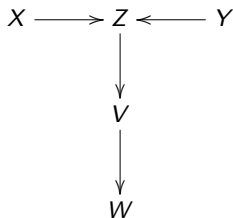
or in a 'fork'

$$X \longleftarrow Z \longrightarrow Y$$

and we have controlled for Z

Rule 2

- A path is blocked if somewhere along the path there is a variable Z that sits in an 'inverted fork'



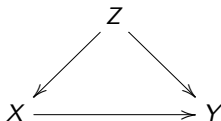
and we have **not** controlled for Z , or any other variable affected by Z .

- Such a variable is called a **collider**
- Notably, even if we control only for W , which is a second-order descendant of the original collider Z , we still capture some of the information contained in Z .
- Therefore, controlling for a descendant of Z will also induce collider bias; However, this bias is likely to be less severe, provided that the descendant is not a deterministic function of Z .

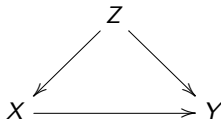
$$X \longleftarrow V \longrightarrow W \longleftarrow Y$$

- Controlling for V blocks the path from X to Y (rule 1)
- Controlling for W leaves the path open (rule 2)
- Controlling for both V and W blocks the path

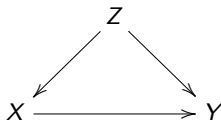
- If all paths between X and Y are blocked, then X and Y are independent
- If at least one path is open between X and Y , then X and Y are generally associated



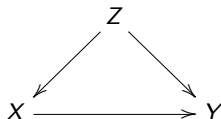
- Suppose that the DAG above depicts the true causal structure
- We want to test whether there is a causal effect of X on Y
 - i.e. does the causal path $X \rightarrow Y$ exist?
- *Control or not control for Z ?*



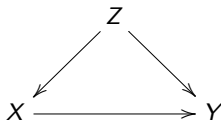
- X = smoking during pregnancy, Y = malformation in the child, Z = mother's age
- Young mothers smoke more often than old mothers, but have lower risk for malformation in their children
- Hence, smokers are more likely to be young than non-smokers, and are for this reason less likely to have children with malformation
- By not controlling for age we may observe a negative association between smoking and malformation, even in the absence of a causal effect



- Suppose that we don't control for Z , and that we observe an association between X and Y
- There are two explanations for this association:
 - the causal path $X \rightarrow Y$
 - the open non-causal path $X \leftarrow Z \rightarrow Y$ (Rule 1)
- Hence, an association between X and Y , when not controlling for Z , does not prove that the causal path $X \rightarrow Y$ exists



- Suppose that we control for Z
 - we block the non-causal path $X \leftarrow Z \rightarrow Y$ (Rule 1)
- Suppose that we then observe an association between X and Y
 - this can only be explained by the causal path $X \rightarrow Y$
- Hence, an association between X and Y , when controlling for Z , proves that there is a causal effect of X on Y



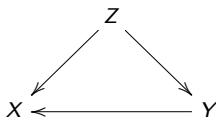
- If the aim is to test for a causal effect of X on Y , then we should control for Z
- We don't have unconditional exchangeability

$$(Y_0, Y_1) \not\perp\!\!\!\perp X$$

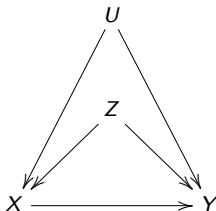
but we have conditional exchangeability, given Z

$$(Y_0, Y_1) \perp\!\!\!\perp X \mid Z$$

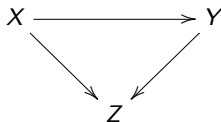
- Controlling for Z may not give a causal effect if the DAG is incorrect, e.g. if
 - Y causes X



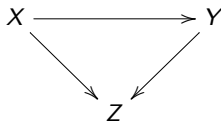
- there are additional common causes of X and Y



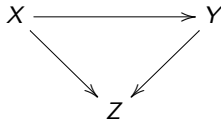
Example



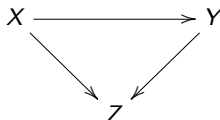
- Suppose that the DAG above depicts the true causal structure
- We want to test whether there is a causal effect of X on Y
 - i.e. does the causal path $X \rightarrow Y$ exist?
- *Control or not control for Z ?*



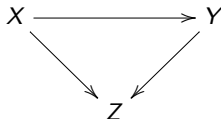
- X = smoking during pregnancy, Y = malformation in the child, Z = birth status (live/stillborn)
- Smoking and malformation increase the risk for stillbirth
- One way of controlling for birth status is by stratification
- Consider the stratum of woman who had stillbirths: **what caused the children to die?**



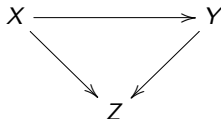
- For the non-smokers, smoking was obviously not the cause
 - perhaps malformation then?
- When smoking is ruled out as the cause of death, the likelihood of malformation increases
 - a negative non-causal association between smoking and malformation!
- By controlling for (e.g. stratifying on) birth status we may observe a negative association between smoking and malformation, even in the absence of a causal effect



- Suppose that we control for Z , and that we observe an association between X and Y
- There are two explanations for this association:
 - the causal path $X \rightarrow Y$
 - the open non-causal path $X \rightarrow Z \leftarrow Y$ (Rule 2)
- Hence, an association between X and Y , when controlling for Z , does not prove that the causal path $X \rightarrow Y$ exists



- Suppose that we don't control for Z
 - we block the non-causal path $X \rightarrow Z \leftarrow Y$ (Rule 2)
- Suppose that we then observe an association between X and Y
 - this can only be explained by the causal path $X \rightarrow Y$
- Hence, an association between X and Y , when not controlling for Z , proves that there is a causal effect of X on Y



- If the aim is to test for a causal effect of X on Y , then we should not control for Z
- We don't have conditional exchangeability, given Z

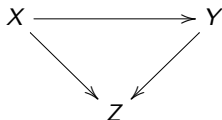
$$(Y_0, Y_1) \not\perp\!\!\!\perp X \mid Z$$

but we have unconditional exchangeability

$$(Y_0, Y_1) \perp\!\!\!\perp X$$

- Control for covariates that block non-causal paths between the exposure and the outcome if controlled for
- Don't control for covariates that open non-causal paths between the exposure and the outcome if controlled for
- If we manage to block all non-causal paths, then any observed association must be due to a causal effect

- If we manage to block all non-causal paths, then any observed association must be due to a causal effect
- We thus have a valid hypothesis test for causation
- Typically, **but not always**, this implies that we have a valid estimate of the causal effect



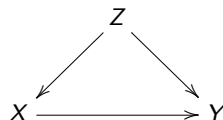
- In left DAG we have (unconditional) exchangeability:

$$(Y_0, Y_1) \perp\!\!\!\perp X$$

- Not controlling for Z gives a valid estimate of the causal effect, as well as a valid test for causation
- In right DAG we have conditional exchangeability, given Z:

$$(Y_0, Y_1) \perp\!\!\!\perp X \mid Z$$

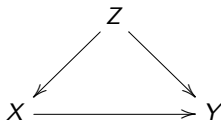
- Controlling for Z gives a valid estimate of the causal effect, as well as a valid test for causation



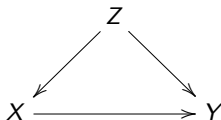
$$X \longrightarrow Y \longrightarrow Z$$

- If we control for Z in the DAG above, then all non-causal paths between X and Y are blocked
 - there are no non-causal paths to start with
- Thus, a conditional association between X and Y , given Z , proves that there is a causal effect of X on Y
 - controlling for Z gives a valid test for causation
- However, it can be shown that we don't have conditional exchangeability, given Z
 - controlling for Z does not give a valid estimate of the causal effect
 - e.g. the conditional risk ratio, given Z , is not equal to the conditional causal risk ratio, given Z

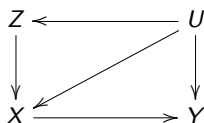
Technical note #2: population causal effects vs subject-specific causal effects



- An arrow from X to Y means that X has a causal effect on Y **for some subjects in the population**
- Typically, but not always, this implies that X has a population causal effect on Y
 - in principle, positive and negative subject-specific causal effects may perfectly cancel out
- More later, when discussing instrumental variables



- When X and Y have common causes, we say that there is **confounding**
- The consequence of confounding is that we don't have exchangeability
- The common causes are called **confounders**
 - now time to modify this definition of 'confounder'

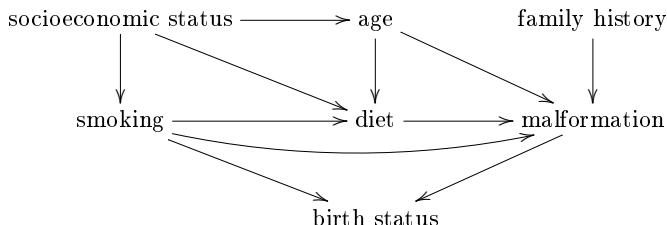


- A **confounder** is a variable that blocks a non-causal path between the exposure and the outcome, if controlled for
 - both Z and U are confounders in the DAG above
- A (set of) variable(s) is **sufficient for confounding control** if the variable(s) blocks all non-causal paths
 - U is sufficient for confounding control, Z is not

$$(Y_0, Y_1) \perp\!\!\!\perp X \mid U$$

$$(Y_0, Y_1) \not\perp\!\!\!\perp X \mid Z$$

A possible DAG for the motivating example

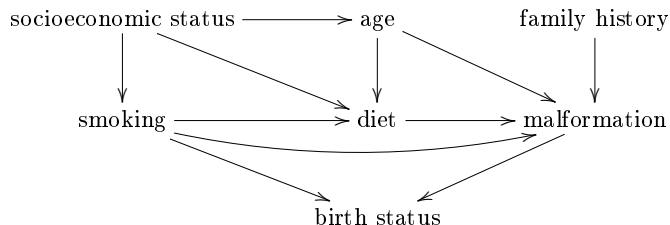


- *Which assumptions are encoded in this DAG?*

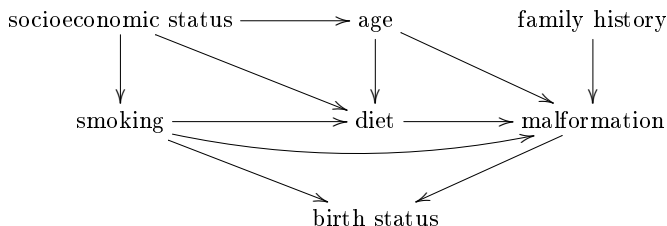
Age has no effect on smoking (although they are statistically dependent), smoking affects (causally) diet, age and smoking has a common cause - socioeconomic status

- *Can these assumptions be tested?*

Yes, to some extent. For example, there is only one open path between age and smoking. Therefore, by controlling for socioeconomic status we can test whether age and smoking are independent (or vice versa - if these variables are dependent when controlling for socioeconomic status it implies that the DAG is incorrect).



- *Given the DAG, which covariates should we control for?*
- *Which covariates would be selected by the data-driven strategies?*



- *Given the DAG, which covariates should we control for?*
socioeconomic status, smoking
- *Which covariates would be selected by the data-driven strategies?*
Will select covariates that are strongly associated with the outcome (malformation). Therefore, such a strategy may select all the covariates above.

- Cannot construct a plausible DAG

socioeconomic status

age

family history

smoking

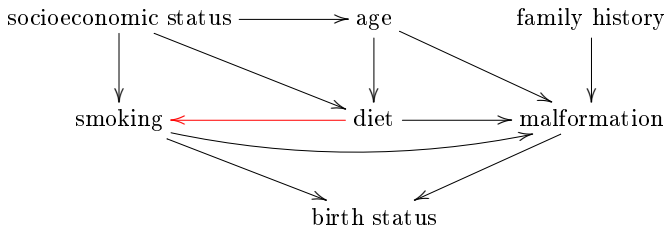
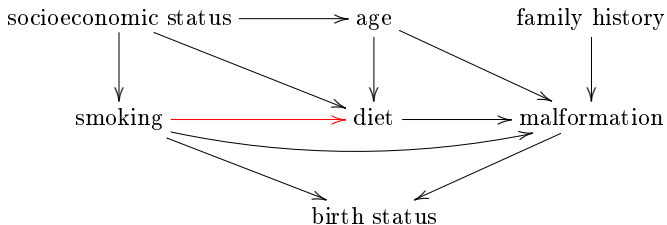
diet

malformation

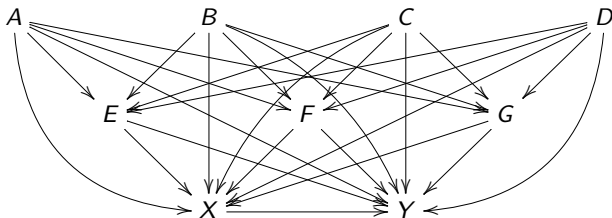
birth status

- DAG-based covariate selection cannot be used, and we have to resort to data-driven strategies
 - but be aware of the pitfalls

- Cannot settle with **one** plausible DAG



- Present all plausible DAGs, and the implied analyses



- All covariates have direct effect on both the exposure and the outcome
- Remember that
 - more covariates requires a bigger model, with a higher potential for bias due to model misspecification
 - some covariates may be prone to measurement errors, and may therefore lead to bias
 - some covariates may reduce statistical power/efficiency when controlled for
- It may be reasonable to exclude some covariates with a weak 'confounding effect'

- Data-driven covariate selection strategies
 - are difficult to apply at the design stage
 - may select non-confounders, which may increase (bias) non-exchangeability if controlled for
- DAGs can be used for knowledge-driven covariate selection
 - encode our *a priori* causal knowledge/beliefs into a DAG
 - apply simple graphical rules to determine what covariates to control for